



Vellum

CS 4900 Capstone by Landon Whitney and
Jackson Jacobson

Login

 **Vellum**

Seed accounts: admin@vellum.test, designer@vellum.test, reviewer@vellum.test

Email

Password

Login

```
function getBearerToken(): string | undefined {
  try {
    if (typeof localStorage !== "undefined") {
      const t = localStorage.getItem(TOKEN_KEY);
      if (t?.trim()) return t.trim();
    }
  } catch {
    // ignore
  }
  return undefined;
}
```

```
async function request<T>(path: string, init?: RequestInit): Promise<T> {
  const headers: Record<string, string> = {
    ...(init?.headers as Record<string, string> | undefined)
  };
}
```

```
const token = getBearerToken();
if (token) {
  headers.Authorization = `Bearer ${token}`;
}
```

```
const response = await fetch(buildUrl(path), {
  ...init,
  headers
});
```

Dashboard/Projects/Queue

 Vellum

DashboardUploadProjectsAssetsManagerAdmin

admin@vellum.testLogout

Review Queue

PROJECTS	ASSETS	NEEDS REVIEW	CHANGES REQUESTED	WIP	APPROVED
6	15	10	2	2	1

Filters

Queue Scope

All Assets

Project

All projects

Search by Title

Search assets

Dashboard Overview

15 matching assets

Asset Queue

NEEDS REVIEW

Map
Ready for reviewer approval
Owner: Admin User
Project: Umbrella Lab Containment Review
Updated: Apr 23, 2026

View Asset

NEEDS REVIEW

README.md
Ready for reviewer approval
Owner: Admin User
Project: Packaging Refresh
Updated: Apr 16, 2026

View Asset

NEEDS REVIEW

Initial Push - Packaging Label QA
Ready for reviewer approval
Owner: Designer User
Project: Spring Launch Campaign
Updated: Apr 16, 2026

View Asset

CHANGES REQUESTED

Initial Push - Landing Page Hero
Needs revision review
Owner: Designer User
Project: Spring Launch Campaign

View Asset

```
router.get("/", async (req, res, next) => {
```

```
  try {
    const uid = parseAuthUserId(req);
    if (uid == null) {
      return res.status(401).json({ error: "Authentication required" });
    }
  }
```

```
  const params = [];
  const whereParts = [];
```

```
  if (!isPlatformAdmin(req.role)) {
    whereParts.push(
      `(
        EXISTS (
          SELECT 1
          FROM organization_members m
          WHERE m.organization_id = p.organization_id
            AND m.user_id = ${params.length + 1}
            AND m.role IN ('OWNER', 'MANAGER')
        )
        OR EXISTS (
          SELECT 1
          FROM project_members pm
          WHERE pm.project_id = p.id
            AND pm.user_id = ${params.length + 1}
        )
      )`
    );
    params.push(uid);
  }
```

```
  const result = await query(
    `SELECT p.id, p.name, p.status, p.organization_id,
      o.name AS organization_name,
      c.name AS client_name,
      (SELECT COUNT(*)::int FROM assets a WHERE a.project_id = p.id) AS asset_count
    FROM projects p
    LEFT JOIN clients c ON c.id = p.client_id
    LEFT JOIN organizations o ON o.id = p.organization_id
    ${whereParts.length ? `WHERE ${whereParts.join(" AND ")}` : ""}
    ORDER BY p.created_at DESC`,
    params
  );

  res.json(result.rows);
} catch (error) {
  next(error);
}
});
```

Asset Detail/Review/Admin

[←](#)

Map

+

-

Shaded

Points

Millions

< 33.40

33.40 - 87.40

87.40 - 151.00

151.00 - 224.00

> 224.00

Enlarge preview

Open preview in new tab

Download file

Owner: Admin User

Project: [Umbrella Lab Containment Review](#)

Current file: [Screenshot 2025-12-10 131459.png](#)

Current version: v1.0

Notes: This is a map of the containment

Status: Needs Review

Review action

Approve (internal) ▾

Apply action

Comments

Versions

Add Comment

Post Comment

No comments yet.



```
export function attachAuth(req, res, next) {
  const header = req.get("Authorization");
  if (!header || !header.startsWith("Bearer ")) {
    req.userId = null;
    req.role = null;
    return next();
  }

  const token = header.slice(7).trim();

  try {
    const payload = verifyAuthToken(token);
    const role = typeof payload.role === "string" ? payload.role.trim().toLowerCase() : "";
    if (!VALID_ROLES.includes(role)) {
      return res.status(401).json({ error: "Invalid token" });
    }

    req.userId = payload.sub !== null ? String(payload.sub) : null;
    req.role = role;
    return next();
  } catch (err) {
    return res.status(401).json({ error: "Invalid or expired token" });
  }
}

export function requireRole(allowedRoles) {
  return (req, res, next) => {
    if (!req.role) {
      return res.status(401).json({ error: "Authentication required" });
    }

    if (!allowedRoles.includes(req.role)) {
      return res.status(403).json({ error: "Forbidden" });
    }

    next();
  };
}
```



Landon - Good, Bad & Ugly

Good:

- The project ended up more complete than we even expected at some points. We created a real full-stack product that is completely working for what we wanted/needed.

Bad:

- Some work took longer than anticipated as we had to revise between sprints. Things like setup/migration was challenging as small factors we didn't consider created errors between our devices we then had to clean up.

Ugly:

- We overcame and conquered, but the actual planning phase between two completely individual schedules and the complications of work and life around it made timelines on some sprints a little more challenging.



Jackson - Good, Bad & Ugly

- **Good:** Backend evolved into a real product model (organizations, projects, assets, role-aware workflows), not just a class demo. RBAC and auth hardening made access rules explicit and safer across API + UI.
- **Bad:** Scope expanded faster than expected, so backend complexity and migration coordination grew a lot.
- **Ugly:** Workflow + permissions + schema changes were tightly coupled, so one small change often touched multiple layers.



Landon - Did we deliver an MVP?

Yes, I would say we have an MVP.

- Users can log in, roles work, and all notable features we outlined are in place
- Core ideas and values are maintained and even elevated beyond a planning stage with specific implementations and changes we made

Stable enough to be released?

- Yes and no? Absolutely stable enough for a product at the scale we designed for a capstone, the code we have in place is well tested and highly functional. There are some factors like file sizes allowed and things alike that would need change for a serious prod release, probably accounting for using a CDN or something alike



Jackson - Did we Deliver an MVP?

- **Yes functionally, we delivered an MVP:** core upload, review, comments, approvals, and role-based access all worked end-to-end. Our application was real, frontend + backend + database were integrated with authenticated API flows.
- **Partially MVP scope grew:** we moved past “minimum” by adding org/project/asset hierarchy and deeper RBAC.
- **Honest takeaway:** we delivered a working MVP, but learned to define tighter “minimum” boundaries earlier.



Website Hosting

- **Frontend:** We deployed the Vite/React app on Vercel as a static site from the frontend folder, with VITE_API_BASE_URL pointing at our API. Live app: <https://cs-4900-frontend.vercel.app/>
- **Backend:** We run the Express API on Render as a Web Service from the backend folder; it serves /api and connects to Postgres using Render's internal DATABASE_URL. API: <https://vellum-backend-unc1.onrender.com/>
- **Database:** We use Render managed PostgreSQL in the same region, synced from our local dev DB with `npm run db:render:sync` (External host for laptop, Internal URL on the web service).

Capstone Expo

We had a great time presenting at UVU Capstone Expo this year we met a lot of awesome people and shared our project. Also got to check out a ton of inspiring projects from other students!





General UVU Student Time Reflection

Landon -

- Coming from SLCC was a little bit of a harsh transition, being taught different languages and coming with less knowledge in some areas, but once I found my footing it was a positive environment.
- (mostly) All of my professors at UVU actually gave me the impression and actions that showed they care about me AND my work, not just my work.

Jackson -

- Coming from Salt Lake Community College was a bit of a tough transition, especially with different languages and gaps in my knowledge, but once I found my footing at Utah Valley University, it became a really positive environment.
- While at UVU, I worked with E2i for two years as a Junior Software Engineer and was involved as VP of the CS Club and Lead of the Google Developer Student Club, which helped me grow both technically and professionally.
- What stood out most was how much my professors genuinely cared about me as a student and not just the work I produced.